STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-SUPPLY-GOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2890-2897 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 2899-2980, the complete author proof (80 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_sdga.json (pb=3102, stmt 2890-2897, proof 2899-2980).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FRB: sources/stacks-sdga/pinned-original-source.tex, source lines 1160–1170. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRX: sources/stacks-sdga/pinned-original-source.tex, source lines 2336–2346. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04J0: sources/stacks-sdga/pinned-original-source.tex, source lines 2980–2980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03CR: sources/stacks-sdga/pinned-original-source.tex, source lines 2980–2980. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $U \in \mathrm{Ob}(\mathcal{C})$. Then $j_!\mathcal{A}_U$ is a good differential graded $\mathcal{A}$-module.*

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $\mathcal{N}$ be a left graded $\mathcal{A}$-module. By Lemma 0FRB we have

$$j_!\mathcal{A}_U \otimes_{\mathcal{A}} \mathcal{N} = j_!(\mathcal{A}_U \otimes_{\mathcal{A}_U} \mathcal{N}|_U) = j_!(\mathcal{N}_U)$$

as graded modules. Since both restriction to U and $j_!$ are exact this proves condition (1). The same argument works for (2) using Lemma 0FRX.

Consider a morphism $(f, f^\sharp) : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ of ringed topoi, a differential graded $\mathcal{O}'$-algebra $\mathcal{A}'$, and a map $\varphi : f^{-1}\mathcal{A} \to \mathcal{A}'$ of differential graded $f^{-1}\mathcal{O}$-algebras. We have to show that

$$f^*j_!\mathcal{A}_U = f^{-1}j_!\mathcal{A}_U \otimes_{f^{-1}\mathcal{A}} \mathcal{A}'$$

satisfies (1) and (2) for the ringed topos $(Sh(\mathcal{C}'), \mathcal{O}')$ endowed with the sheaf of differential graded $\mathcal{O}'$-algebras $\mathcal{A}'$. To prove this we may replace $(Sh(\mathcal{C}), \mathcal{O})$ and $(Sh(\mathcal{C}'), \mathcal{O}')$ by equivalent ringed topoi. Thus by Modules on Sites, Lemma 03CR we may assume that f comes from a morphism of sites $f : \mathcal{C} \to \mathcal{C}'$ given by the continuous functor $u : \mathcal{C} \to \mathcal{C}'$. In this case, set $U' = u(U)$ and denote $j' : Sh(\mathcal{C}'/U') \to Sh(\mathcal{C}')$ the corresponding localization morphism. We obtain a commutative square of morphisms of ringed topoi

$$\begin{array}{ccc} (Sh(\mathcal{C}'/U'), \mathcal{O}'_{U'}) & \xrightarrow{(j',(j')^\sharp)} & (Sh(\mathcal{C}'), \mathcal{O}') \\ \downarrow{\scriptstyle (f',(f')^\sharp)} & & \downarrow{\scriptstyle (f,f^\sharp)} \\ (Sh(\mathcal{C}/U), \mathcal{O}_U) & \xrightarrow{(j,j^\sharp)} & (Sh(\mathcal{C}), \mathcal{O}). \end{array}$$

and we have $f'_*(j')^{-1} = j^{-1}f_*$. See Modules on Sites, Lemma 04J0. By uniqueness of adjoints we obtain $f^{-1}j_! = j'_!(f')^{-1}$. Thus we obtain

$$\begin{aligned} f^*j_!\mathcal{A}_U &= f^{-1}j_!\mathcal{A}_U \otimes_{f^{-1}\mathcal{A}} \mathcal{A}' \\ &= j'_!(f')^{-1}\mathcal{A}_U \otimes_{f^{-1}\mathcal{A}} \mathcal{A}' \\ &= j'_!\left((f')^{-1}\mathcal{A}_U \otimes_{f^{-1}\mathcal{A}|_{U'}} \mathcal{A}'|_{U'}\right) \\ &= j'_!\mathcal{A}'_{U'} \end{aligned}$$

The first equation is the definition of the pullback of $j_!\mathcal{A}_U$ to a differential graded module over $\mathcal{A}'$. The second equation because $f^{-1}j_! = j'_!(f')^{-1}$. The third equation by Lemma 0FRX applied to the ringed site $(\mathcal{C}', f^{-1}\mathcal{O})$ with sheaf of differential graded algebras $f^{-1}\mathcal{A}$ and with differential graded modules $(f')^{-1}\mathcal{A}_U$ on $\mathcal{C}'/U'$ and $\mathcal{A}'$ on $\mathcal{C}'$. The fourth equation holds because of course we have $(f')^{-1}\mathcal{A}_U = f^{-1}\mathcal{A}|_{U'}$. Hence we see that the pullback is another module of the same kind and we've proven conditions (1) and (2) for it above. □